CSC 380 — Midterm Study Guide (PROVISIONAL · LIGHT · date NV)

Banner — date NEEDS VERIFICATION; scope PROVISIONAL; LIGHT guide.

- **Date:** the syllabus says **"tentatively"** Tue Oct 20, 2026, 5:00–6:15 PM Arizona time, in the regular lecture. The calendar also lists Oct 20. The date stays **Needs verification** until the instructor confirms it.
- **Scope:** coverage is **not published**. It is provisional, from syllabus weeks 1–8: pandas, visualization, probability, random variables and distributions, covariance and correlation, and midterm review.
- **Light only:** this is a concept map plus a small practice set. Do not do dense prep that assumes a confirmed date.

Canonical maps: `csc380-topic-map.md` · `announced-assessments-inventory-2026-09-24.md` · `site-audit-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00–6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Midterm (light provisional guide)	Item exists
Date/time	Tentatively Tue Oct 20, 2026, 5:00–6:15 PM Arizona	**Needs verification**
Format	**Not published.** Gradescope delivery is not confirmed.	NV
Coverage	Syllabus weeks 1–8	**PROVISIONAL**
Not yet lectured	Special Distributions (week 6), Covariance/Correlation (week 7), and Midterm Review (week 8) are still ahead	Expected
Guide updated	2026-09-24 (America/Phoenix) — Opus review pass	

All times are **Arizona time (MST)**.

2. Topic checklist (light concept map)

Order	Section ID	Topic (syllabus week — provisional)
1	`380-L-intro`	Course overview (wk 1)
2	`380-L-pandas`	Basic data manipulation with pandas (wk 1)
3	`380-L-viz`	Basic visualization (wk 1)
4	`380-L-prob1`	Probability intro; conditional probability (wk 2)
5	`380-L-comb`	Combinations (wk 3) — *new in this review*
6	`380-Q04-indep`	Independence (wk 3; also Quiz 04)
7	`380-Q04-bayes` / `380-Q04-lotp`	Bayes and LOTP (Quiz 04 **published** subset)
8	`380-L-rv`	Random variables; distributions; discrete vs continuous (wk 4–5)
9	`380-L-special`	Special distributions (wk 6 — as lectured)
10	`380-L-corr`	Covariance and correlation (wk 7 — as lectured)
11	`380-L-review`	Midterm review (wk 8)

Do **not** start the post-midterm material (modeling, regression, classification).

3. Mini explanations

- **pandas.** These are standard pandas; match your class notebooks.
- Filter rows: `df[df["age"] > 30]`
- Select columns: `df[["a", "b"]]`
- Group means: `df.groupby("team")["score"].mean()`
- Missing values: `df.isna().sum()` and `df.dropna()`
- **Chart choice.**

What you want to show	Chart
Distribution of one numeric variable	Histogram or box plot
Two numeric variables	Scatter plot
Comparison across categories	Bar chart
Change over time	Line chart

- **Combinations.** C(n, k) = n! / (k!(n−k)!) counts **unordered** selections. Permutations P(n, k) = n! / (n−k)! count **ordered** ones.
- **Random variables.** For a discrete X, $E[X] = \Sigma\, x \cdot P(x)$. $Var(X) = E[X^2] - (E[X])^2$.
- **Special distributions** (as lectured).
- Bernoulli(p): mean p, variance p(1−p).
- Binomial(n, p): mean np, variance np(1−p).
- Uniform, Poisson, Normal, and Exponential follow if covered.
- **Covariance and correlation.** $Cov(X, Y) = E[(X - \mu X)(Y - \mu Y)]$. Corr = Cov / (σX·σY), which always lies in [−1, 1]. Correlation measures **linear** association only. Check which denominator the course uses for sample covariance (n or n − 1).

- **Bayes, LOTP, independence.** These are the published Quiz 04 topics; see the Quiz 04 Practice Guide.

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4. Light practice (self-contained)

L1 (recall). Define independence, and write correlation in terms of covariance.

L2 (combinations). How many 3-person committees can be chosen from 7 people? What is the probability of exactly 2 heads in 4 fair coin flips?

L3 (random variable). For a fair six-sided die X, find E[X] and Var(X).

L4 (binomial). X ~ Binomial(10, 0.3). Find E[X] and Var(X).

L5 (covariance/correlation). For X = [1, 2, 3] and Y = [2, 4, 6], find the sample covariance (denominator $n - 1$) and the correlation.

L6 (Bayes). P(H) = 0.2, P(E | H) = 0.9, P(E | not-H) = 0.3. Find P(H | E).

L7 (independence). P(A) = 0.25, P(B) = 0.4, P(A and B) = 0.1. Are A and B independent?

L8 (visualization). Pick a chart for (a) the distribution of house prices, (b) price vs. square footage, (c) average price by neighborhood.

L9 (pandas). Write one line that gives the mean `score` for each `team` in DataFrame `df`.

L10 (planning). Why must the tentative Oct 20 date be re-verified before making a dense plan?

4b. Format reminder — Practice recommendation

The Gradescope format notes show that quizzes are about 1.5 points on a single page, and each homework has **two** items (written and Code). The **midterm format is not published**. For practice, write L2–L7 by hand, one per half-page, with every step shown. This is a generic habit, not a claimed midterm format.

5. ANSWER KEY

L1. Independence means P(A and B) = P(A)·P(B). Corr(X, Y) = Cov(X, Y) / (σX·σY).

L2. C(7, 3) = 35. P(exactly 2 heads in 4 flips) = C(4, 2)/2^4 = 6/16 = **0.375**.

L3. E[X] = 3.5. E[X^2] = 91/6. Var(X) = 91/6 − 12.25 = **35/12 ≈ 2.917**.

L4. E[X] = 10·0.3 = **3**. Var(X) = 10·0.3·0.7 = **2.1**.

L5. The means are 2 and 4. The deviations are (−1, 0, 1) for X and (−2, 0, 2) for Y, so the sum of products is 2 + 0 + 2 = 4. Sample covariance = 4 / (3 − 1) = **2**. The sample standard deviations are 1 and 2, so correlation = 2 / (1·2) = **1**, a perfect positive linear relationship. With the population denominator, the covariance is 4/3, and the correlation is still 1.

L6. P(E) = 0.9·0.2 + 0.3·0.8 = 0.42. P(H | E) = 0.18 / 0.42 = **3/7 ≈ 0.429**.

L7. 0.25 × 0.4 = 0.1 = P(A and B). **Yes, independent.**

L8. (a) Histogram (or box plot). (b) Scatter plot. (c) Bar chart.

L9. `df.groupby("team")["score"].mean()`

L10. The syllabus marks the date as "tentative". The context rule is never to treat an unconfirmed date as fixed, so re-check on D2L, the course site, or in class during Oct 13–19 before planning around it.

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6. Common mistakes + readiness (light)

- Studying regression or neural networks early.
- Treating the tentative Oct 20 date as fixed.
- Skipping Quiz 04 fluency, since Bayes, LOTP, and independence are the only **published** subset.
- Mixing up combinations (unordered) with permutations (ordered).
- Reading a correlation near 0 as "no relationship", when it only means no **linear** relationship.
- Inventing quiz dates from the Tuesday cadence.

Ready (light):

- [ ] Weeks 1–8 concept map reviewed
- [ ] Quiz 04 topics solid
- [ ] L1–L10 correct
- [ ] Date and scope re-verified during Oct 13–19 **before** densifying

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7. Prep plan (light — no day-before cram that assumes a confirmed date)

Window	Action
Now – early Oct	Quiz 04 practice and HW3/HW4 probability. **Not** a midterm marathon.
Sat Oct 3	One light pass over the `380-L-*` concept map and L1–L10
Tue Oct 13	HW4 due 11:59 PM (written plus Code)
Oct 13–19	**Re-verify** the date and scope on D2L, the site, or the tracker. Densify only after that.
If Oct 20 is confirmed	Sit the midterm at 5:00–6:15 PM. The night before, calm review only.

Do deeper work Mon/Wed/Fri and weekends; keep Tuesdays and Thursdays light.

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8. Changes in this review (2026-09-24)

- Kept the **date NV** and **LIGHT** banners unchanged in meaning. Changed "PT" to Arizona time.
- **Made the answer key self-contained.** The old key said "see the Quiz 04 guide" for Bayes; all L-items now have worked answers.
- Added the week-3 **Combinations** topic, which was merged into independence and had no practice. Added practice on random variables, the binomial distribution, covariance/correlation, visualization, and pandas (L2–L5, L8–L9).
- Noted that the weeks 6–8 lectures have not happened yet.
- Kept the new practice to ten short items so the guide stays light.

9. Disclaimer

Study aid only — not for submitted work. Follow CSC 380 and UA policies. The title marks provisional/NV status on purpose.